Transport in a constant type family: for P(x) equal to a fixed B there is a path from the transport of b along p to b

The Univalent Foundations Program

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Editorial note. The counted statement is the author lemma labelled thm:trans-trivial (source lines 896-899); the proof span 900-910 is the author's first proof environment following it, retained line for line.

Editorial note. The statement itself introduces the transconst notation used by the later lemma thm:apd-const.

Editorial note. The author's second proof at the following lines is not counted and not retained on the card face.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: transport in a constant type family is trivial up to a path. Only the counted lemma and its first author proof are retained; the author's second proof remains uncounted in the source.

Allowed standard prerequisites (editorial).

- Dependent type families over A and the identity type of A with its reflexivity element.
- Transport along a path: for p in x = y a function from P(x) to P(y), with the computation rule that transport along refl x is the identity function by definition.
- The induction principle for identity types (path induction) with its conversion rule.

Lemma 0.0.1. *If $P : A \to \mathcal{U}$ is defined by $P(x) :\equiv B$ for a fixed $B : \mathcal{U}$, then for any $x, y : A$ and $p : x = y$ and $b : B$ we have a path*

$$\mathsf{transportconst}_p^B(b) : \mathsf{transport}^P(p, b) = b.$$

First proof. Fix a $b : B$, and let $D : \prod_{(x,y:A)}(x = y) \to \mathcal{U}$ be the type family defined by

$$D(x, y, p) :\equiv (\mathsf{transport}^P(p, b) = b).$$

Then $D(x, x, \mathsf{refl}_x)$ is $(\mathsf{transport}^P(\mathsf{refl}_x, b) = b)$, which is judgmentally equal to $(b = b)$ by the computation rule for transporting. Thus, we have the function

$$d :\equiv \lambda x.\, \mathsf{refl}_b : \prod_{x:A} D(x, x, \mathsf{refl}_x).$$

Now path induction gives us an element of $\prod_{(x,y:A)} \prod_{(p:x=y)} (\mathsf{transport}^P(p,b) = b)$, as desired. □